Production–decay scale correlations and width conventions in fiducial trileptonic $t\bar{t}W^+$ production

Updated comparison from forty completed integrations

ttW production–decay study

30 October 2026

Abstract

We combine ten repeated integrations for each of four predictions: additive (S) or multiplicative (Π) NLO QCD production and top decays, with scale-dependent or fixed total widths. The samples use the updated common lepton selection, applied during generation, and retain 581 histograms and all 81 production/decay scale combinations. The selected total rates agree at the 0.13% level, while Π increases the exclusive zero- and one-bottom-jet rates by approximately 16.8% and 7.8%, and decreases the two-tag rate by 4.0%. For the two-tag multiplicative prediction with fixed widths, the scale envelope is $^{+7.30}_{-7.85}\%$ with correlated variations and $^{+14.01}_{-13.29}\%$ with independent production and decay scales. The combined native Monte Carlo error on the selected total is about 0.05%; repeated-run scatter provides an additional precision diagnostic and permits paired statistical error estimates on ratios. We present fiducial lepton, jet, invariant-mass and angular distributions, distinguish scale-prescription effects from integration errors, and discuss the relation to published full off-shell and higher-order calculations. The process is one direct-lepton assignment with W^+ at 13 TeV.

1 Campaign inputs and fiducial definition

The simulated process is

$$pp \to t\bar{t}W^+ \to b\,e^+\nu_e\;\bar{b}\,e^-\bar{\nu}_e\;\mu^+\nu_\mu + X, \qquad \sqrt{s} = 13\ \text{TeV}. \tag{1}$$

Production and both top decays include NLO QCD corrections with spin correlations. The three leptonic W decays are LO, all resonances are on shell, and $m_b = 0$. We use the local fNLO implementation in MadGraph5_aMC@NLO and FKS subtraction[1, 2]. The real-diagram production hard-channel split uses $h = 0.02$; this is a sampling parameter, not a physical cut on emitted radiation.

This update uses `Events/run_01` through `Events/run_10` in each of the following process directories:

Prediction	Directory	Integrations
S, varying widths	`ttw_additive`	10
Π, varying widths	`ttw_multiplicative`	10
S, fixed widths	`ttw_additive_fixedwidth`	10
Π, fixed widths	`ttw_multiplicative_fixedwidth`	10

All forty final HwU files are complete and have the same bin and scale layouts. Archived cards define each input and are copied into this report. The parameter cards agree exactly. Installed analysis and cut sources also agree across the four directories and are copied into the report inputs with their hashes. These installed source snapshots supplement the run archives, which themselves contain cards rather than analysis source. The native four-stage cutflows have identical entries in every stage, as expected when the complete baseline is already imposed during generation. The earlier report of 19 October and its event files retain their older selections; none of those integrations is included here. No new integrations were launched to prepare this update.

The inputs use $m_t = 172.5$ GeV, $m_W = 80.385$ GeV, and the central member of NNPDF40_nlo_as_01180 (LHAPDF ID 331700). The reference scales are

$$\mu^P_{R,0} = \mu^P_{F,0} = \frac{1}{2} H^P_T, \qquad \mu^t_{R,0} = \mu^{\bar{t}}_{R,0} = m_t. \tag{2}$$

The production H^P_T sums transverse masses of the undecayed top, antitop and associated W, plus the transverse momenta of production radiation. The associated W is grouped at its physical invariant mass; decay radiation is not added to the production scale. Each integration requested `req_acc_FO=0.002`. Scale reweighting is enabled and PDF reweighting is disabled. The accuracy target concerns an integrated rate, not every histogram bin. The saved timing summaries report about 526 aggregate worker-days for the campaign, including the integration stages; this is summed worker time, not elapsed wall time.

Every charged lepton must satisfy

$$p_{\mathrm{T}}(\ell) \geq 20 \text{ GeV}, \qquad |\eta(\ell)| < 2.5. \tag{3}$$

The opposite-sign electron pair must have $m_{ee} > 12$ GeV. There are no dilepton or three-lepton Z-mass vetoes. All QCD partons are clustered with anti-k_T, $R = 0.4$, and four-vector recombination. Accepted jets have $p_{\mathrm{T}} \geq 25$ GeV and $|\eta| < 2.5$. Each lepton must be separated from every accepted jet by

$$\Delta R(\ell, j) \geq \min\left(0.4,\ 0.04 + \frac{10 \text{ GeV}}{p_{\mathrm{T}}(\ell)}\right). \tag{4}$$

There is no missing-momentum cut and no minimum jet or bottom-tag count in this baseline. Bottom jets participate in isolation as ordinary jets. The shared baseline helper acts on each full lab-frame real or subtraction event before the generic decay-product mask. It therefore remains active with the archived `cut_decays=False` setting.

The histogram labelled "inclusive rate" is now the baseline-selected total. It is identical to the final cutflow bin and is not an unrestricted production cross section. We suppress that duplicate entry in the tables. The exclusive $0b$, $1b$ and $2b$ rates sum to this selected total separately at every scale point, within printed-output rounding. A tagged jet contains the primary b or primary antibottom but not both. A merged b–antibottom jet is untagged and remains an ordinary jet for isolation and activity. Species and pair observables are filled only when their required accepted objects exist. This definition concerns the supported final state with one primary b and one primary antibottom; it is not a general flavour algorithm for arbitrary additional bottom pairs.

2 Perturbative and width conventions

Write the production density as $B_P + \Delta_P$ and a top-decay density as $D^{(0)}_i + D^{(1)}_i$. The symbols Δ_P and (1) denote complete NLO corrections, including real radiation, virtual terms and subtraction. With $g_i = \Gamma^{(1)}_i / \Gamma^{(0)}_i$, the strict additive result is schematically

$$\begin{aligned} d\sigma_S = (B_P + \Delta_P) \otimes \frac{D^{(0)}_t}{\Gamma^{(0)}_t} \otimes \frac{D^{(0)}_{\bar{t}}}{\Gamma^{(0)}_{\bar{t}}} \\ + B_P \otimes \left[\frac{D^{(1)}_t - g_t D^{(0)}_t}{\Gamma^{(0)}_t} \otimes \frac{D^{(0)}_{\bar{t}}}{\Gamma^{(0)}_{\bar{t}}} + \frac{D^{(0)}_t}{\Gamma^{(0)}_t} \otimes \frac{D^{(1)}_{\bar{t}} - g_{\bar{t}} D^{(0)}_{\bar{t}}}{\Gamma^{(0)}_{\bar{t}}} \right]. \end{aligned} \tag{5}$$

The multiplicative result retains products of corrected densities,

$$d\sigma_{\Pi} = (B_P + \Delta_P) \otimes \frac{D^{(0)}_t + D^{(1)}_t}{\Gamma^{\mathrm{NLO}}_t} \otimes \frac{D^{(0)}_{\bar{t}} + D^{(1)}_{\bar{t}}}{\Gamma^{\mathrm{NLO}}_{\bar{t}}}. \tag{6}$$

Common LO leptonic W factors are suppressed. The products include spin contractions and phase-space integration, not multiplication of spin-averaged rates. Both prescriptions have the same strict NLO

expansion. Their difference probes selected higher-order terms, not a complete NNLO correction. The present Π prescription retains production–decay and decay–decay products in addition to unexpanded normalizations; it is not automatically the same as a literature result described only as an unexpanded NWA.

Both width prescriptions use the same nominal values,

$$\Gamma_t^{\mathrm{LO}} = 1.48062850925\ \mathrm{GeV}, \quad \Gamma_t^{\mathrm{NLO}} = 1.35354852119\ \mathrm{GeV}, \quad \Gamma_W^{\mathrm{LO}} = 2.09767356205\ \mathrm{GeV}. \tag{7}$$

With `AUTO`, the NLO top width varies as

$$\Gamma_t^{\mathrm{NLO}}(\xi m_t) = \Gamma_t^{\mathrm{LO}} + [\Gamma_t^{\mathrm{NLO}}(m_t) - \Gamma_t^{\mathrm{LO}}]\frac{\alpha_s(\xi m_t)}{\alpha_s(m_t)}. \tag{8}$$

The fixed-width samples instead use `EXPLICIT` with the same LO and NLO widths at factors 1/2, 1 and 2. Positive-PDG card inputs cover both top charges, although the two decay scale axes remain separate. The W width is constant in both cases because its leptonic decay is LO. In S, freezing the widths also freezes the width-expansion counterterm; in Π, it freezes the NLO normalization denominators. No Breit–Wigner kinematics is introduced by either choice.

The two width conventions have identical central definitions within each approximation. Their small measured central differences arise from the integrations, not a physical width change. The normalized decay response can largely cancel when numerator and denominator vary together. At fixed width that normalization cancellation is absent: lowering a decay scale strengthens the negative decay correction, whereas lowering the production scale generally increases the rate. Correlated variations can therefore compensate. Fiducial acceptance prevents the unrestricted inclusive normalization identities from being applied directly to the selected total in this report.

3 Replica combination and uncertainty definitions

For each observable bin and each scale vector, we take an equal-weight mean of the $K = 10$ completed estimates,

$$\bar{x} = \frac{1}{K}\sum_{k=1}^{K} x_k, \qquad e_{\mathrm{native}} = \frac{1}{K}\sqrt{\sum_{k=1}^{K} e_k^2}, \qquad e_{\mathrm{replica}} = \sqrt{\frac{1}{K(K-1)}\sum_{k=1}^{K}(x_k - \bar{x})^2}. \tag{9}$$

The central value does not use estimated inverse-variance weights: this avoids favouring a replica simply because it happened to estimate a small variance. Absolute tables and plot bars use the propagated native errors. The independent estimate from replica scatter is reported separately in Table 4 and Fig. 10. Neither prescription certifies unsampled tails.

The logged Ranmar seed pairs are distinct between different run-index blocks, including across the four directories. We treat these ten blocks as independent repeated integrations. Within a matching run index, the predictions share streams: the logged S/Π overlaps are 747 and 745 distinct pairs for varying and fixed widths, respectively. The width partners share still more streams. Thus statistical errors between predictions must not be combined as independent errors. The repeated blocks now permit an empirical estimate of this covariance.

For the central ratio $\widehat{R} = \bar{P}/\bar{S}$, the plotted statistical error is the paired delete-one jackknife standard error,

$$R_{(-k)} = \frac{\sum_{j\neq k} P_j}{\sum_{j\neq k} S_j}, \qquad e_R = \sqrt{\frac{K-1}{K}\sum_{k=1}^{K}\left(R_{(-k)} - \overline{R}_{(-)}\right)^2}. \tag{10}$$

Matching indices are omitted together, preserving their observed correlation. This is a finite-ten-replica precision estimate, not a claim of exact Gaussian coverage or a substitute for more replicas in rare tails. For fixed-minus-varying central differences we similarly use the scatter of paired differences. Scale-weight

MC covariances inside one integration were not saved; no statistical error is assigned to the scale-envelope boundaries themselves.

A scale point is ordered as $\boldsymbol{\xi} = (\xi_R, \xi_F, \xi_t, \xi_{\bar{t}})$, with each factor in $\{1/2, 1, 2\}$. Let $\mathcal{P}_7$ contain the production pairs with $1/2 \leq \xi_R/\xi_F \leq 2$. The reductions are

Name	Retained points	Count
Production7	$(r, f, 1, 1)$, $(r, f) \in \mathcal{P}_7$	7
Decay3	$(1, 1, d, d)$	3
Decay9	$(1, 1, t, a)$	9
Common3	(d, d, d, d)	3
C7: correlated	(r, f, r, r), $(r, f) \in \mathcal{P}_7$	7
I21: independent production/decay	(r, f, d, d), $(r, f) \in \mathcal{P}_7$	21
I63: independent production and both decays	(r, f, t, a), $(r, f) \in \mathcal{P}_7$	63
Full81 diagnostic	(r, f, t, a) with all production pairs	81

C7 ties decay renormalization factors to production renormalization; factorization still has its permitted seven-point choices. Common3 also ties factorization. I21 separates production from a common decay factor; I63 additionally separates the top and antitop. Independence here describes the allowed scale coordinates, not a probability model or a quadrature prescription for theory errors.

Scale envelopes are reduced *after* averaging replica weights. For ratios, each four-component vector is shared between numerator and denominator before taking minima or maxima,

$$R_{\min,\max} = \min_{\boldsymbol{\xi}\in\mathcal{G}}, \max \frac{\overline{d\sigma_\Pi}(\boldsymbol{\xi})}{\overline{d\sigma_S}(\boldsymbol{\xi})}. \tag{11}$$

Dividing unrelated absolute-envelope extrema would answer a different question. Common scale choices *between* the two approximations are distinct from correlated production/decay choices *within* one approximation. Bands are sensitivity estimates, not confidence intervals or guaranteed bounds on missing corrections.

4 Selected rates, scale responses and integration precision

Width	Approx.	Baseline	$0b$	$1b$	$2b$
Varying	S	0.237181 ± 0.000111	0.009584 ± 0.000024	0.060965 ± 0.000071	0.166631 ± 0.000088
Varying	Π	0.236888 ± 0.000117	0.011196 ± 0.000078	0.065740 ± 0.000115	0.159953 ± 0.000118
Fixed	S	0.237190 ± 0.000111	0.009585 ± 0.000024	0.060968 ± 0.000071	0.166637 ± 0.000088
Fixed	Π	0.236892 ± 0.000117	0.011193 ± 0.000078	0.065739 ± 0.000115	0.159960 ± 0.000118

Table 1: Equal-weight mean cross sections in fb, with propagated native MC errors. The baseline is the generation-selected total; the three exclusive tag categories partition it.

The selected totals are 0.237181 and 0.236888 fb for S and Π with varying widths, and 0.237190 and 0.236892 fb with fixed widths. The ratio shifts are only -0.123% and -0.126%, with empirical paired standard errors of about 0.083 and 0.084 percentage points. These data support close agreement of the total; they do not resolve a large total normalization correction.

The exclusive categories behave differently. The zero-tag enhancement is 16.81% with varying widths and 16.77% with fixed widths. The one-tag enhancement is 7.83% in both cases, while the two-tag decrease is 4.01%. Their paired MC errors on the percentage shifts are approximately 0.95, 0.20 and 0.10 percentage points, respectively. The pattern is an acceptance-dependent redistribution among tag categories. A small total difference cannot establish equivalence of the fiducial shapes or category efficiencies.

All three category ratios remain on the same side of unity throughout their I63 sets. Absolute scale bands can nevertheless overlap because large common responses cancel in the scale-matched ratio. This quantifies the difference between the two prescriptions; it does not turn either one into a complete

Width	Quantity	Baseline	$0b$	$1b$	$2b$
Varying	$100(\Pi/S-1)$	-0.12	$+16.81$	$+7.83$	-4.01
	Paired MC error on Π/S	0.00083	0.00951	0.00200	0.00097
	C7 ratio range	$[0.9987, 0.9990]$	$[1.1636, 1.1726]$	$[1.0643, 1.0860]$	$[0.9586, 0.9638]$
	I63 ratio range	$[0.9980, 0.9996]$	$[1.1246, 1.2153]$	$[1.0550, 1.1012]$	$[0.9475, 0.9721]$
Fixed	$100(\Pi/S-1)$	-0.13	$+16.77$	$+7.83$	-4.01
	Paired MC error on Π/S	0.00084	0.00958	0.00208	0.00099
	C7 ratio range	$[0.9927, 1.0057]$	$[1.1521, 1.1772]$	$[1.0552, 1.0949]$	$[0.9584, 0.9653]$
	I63 ratio range	$[0.9895, 1.0057]$	$[1.1325, 1.2018]$	$[1.0552, 1.0949]$	$[0.9403, 0.9762]$

Table 2: Central percentage shifts $100(\Pi/S-1)$, paired statistical errors on the dimensionless ratio, and scale-matched ratio envelopes. Only the first row of each width block is expressed in percent.

higher-order prediction. The selected total must not be used as the unrestricted inclusive normalization check discussed for the previous, unfiltered integrations.

Width	Approx.	Selection	C7	I21	I63	Full81
Varying	S	Baseline	$^{+10.70}_{-10.20}\%$	$^{+10.78}_{-10.26}\%$	$^{+10.78}_{-10.26}\%$	$^{+10.78}_{-10.26}\%$
Varying	S	$0b$	$^{+16.15}_{-13.35}\%$	$^{+16.15}_{-13.35}\%$	$^{+16.15}_{-13.35}\%$	$^{+16.15}_{-13.35}\%$
Varying	S	$1b$	$^{+14.46}_{-12.41}\%$	$^{+14.46}_{-12.41}\%$	$^{+14.46}_{-12.41}\%$	$^{+14.46}_{-12.41}\%$
Varying	S	$2b$	$^{+9.01}_{-9.21}\%$	$^{+10.77}_{-10.39}\%$	$^{+10.77}_{-10.39}\%$	$^{+10.77}_{-10.39}\%$
Varying	Π	Baseline	$^{+10.72}_{-10.21}\%$	$^{+10.87}_{-10.32}\%$	$^{+10.87}_{-10.32}\%$	$^{+10.87}_{-10.32}\%$
Varying	Π	$0b$	$^{+15.70}_{-13.40}\%$	$^{+15.70}_{-13.40}\%$	$^{+15.70}_{-13.40}\%$	$^{+15.70}_{-13.40}\%$
Varying	Π	$1b$	$^{+12.98}_{-11.79}\%$	$^{+12.98}_{-11.79}\%$	$^{+12.98}_{-11.79}\%$	$^{+12.98}_{-11.79}\%$
Varying	Π	$2b$	$^{+9.45}_{-9.34}\%$	$^{+12.17}_{-11.55}\%$	$^{+12.17}_{-11.55}\%$	$^{+12.17}_{-11.55}\%$
Fixed	S	Baseline	$^{+9.19}_{-9.37}\%$	$^{+12.03}_{-11.27}\%$	$^{+12.03}_{-11.27}\%$	$^{+12.03}_{-11.27}\%$
Fixed	S	$0b$	$^{+14.94}_{-12.69}\%$	$^{+14.94}_{-12.69}\%$	$^{+14.94}_{-12.69}\%$	$^{+14.94}_{-12.69}\%$
Fixed	S	$1b$	$^{+13.18}_{-11.71}\%$	$^{+13.18}_{-11.71}\%$	$^{+13.18}_{-11.71}\%$	$^{+13.18}_{-11.71}\%$
Fixed	S	$2b$	$^{+7.40}_{-8.32}\%$	$^{+12.11}_{-11.48}\%$	$^{+12.11}_{-11.48}\%$	$^{+12.11}_{-11.48}\%$
Fixed	Π	Baseline	$^{+8.55}_{-8.74}\%$	$^{+12.68}_{-12.09}\%$	$^{+12.68}_{-12.09}\%$	$^{+12.68}_{-12.09}\%$
Fixed	Π	$0b$	$^{+13.41}_{-11.98}\%$	$^{+13.41}_{-11.98}\%$	$^{+13.41}_{-11.98}\%$	$^{+13.41}_{-11.98}\%$
Fixed	Π	$1b$	$^{+10.75}_{-10.35}\%$	$^{+10.75}_{-10.35}\%$	$^{+10.75}_{-10.35}\%$	$^{+10.75}_{-10.35}\%$
Fixed	Π	$2b$	$^{+7.30}_{-7.85}\%$	$^{+14.01}_{-13.29}\%$	$^{+14.01}_{-13.29}\%$	$^{+14.01}_{-13.29}\%$

Table 3: Scale envelopes relative to each mean central rate, in percent. Full81 additionally includes the extreme production μ_R/μ_F ratios excluded from the usual seven-point set.

For the two-tag rate, varying widths give $S:\ ^{+9.02}_{-9.21}\%$ (C7) versus $^{+10.77}_{-10.39}\%$ (I63), and $\Pi:\ ^{+9.45}_{-9.33}\%$ versus $^{+12.18}_{-11.55}\%$. With fixed widths the contrast grows: $S:\ ^{+7.40}_{-8.32}\%$ versus $^{+12.11}_{-11.48}\%$, and $\Pi:\ ^{+7.30}_{-7.85}\%$ versus $^{+14.01}_{-13.29}\%$. The response slices in Fig. 3 show the opposing production and decay slopes. The correlated fixed-width band samples a direction with partial compensation; it is not evidence for a corresponding improvement in perturbative accuracy.

The I21 and I63 extrema coincide for these integrated rates, and adding the remaining Full81 points does not enlarge their extrema. In the zero- and one-tag categories even C7 reaches the same extrema as I63. These are numerical properties of the present integrated responses, not identities for differential bins. The full stored grid is retained.

The propagated native error on the selected total is 0.047–0.049%; the empirical replica error is 0.049–

Width	Approx.	Selection	Native MC [%]	Replica SE [%]	Replica/native
Varying	S	Baseline	0.047	0.056	1.20
Varying	S	$0b$	0.252	0.325	1.29
Varying	S	$1b$	0.117	0.127	1.09
Varying	S	$2b$	0.053	0.061	1.14
Varying	Π	Baseline	0.049	0.049	0.99
Varying	Π	$0b$	0.696	0.717	1.03
Varying	Π	$1b$	0.175	0.107	0.61
Varying	Π	$2b$	0.074	0.070	0.95
Fixed	S	Baseline	0.047	0.057	1.22
Fixed	S	$0b$	0.253	0.337	1.33
Fixed	S	$1b$	0.117	0.129	1.11
Fixed	S	$2b$	0.053	0.061	1.14
Fixed	Π	Baseline	0.049	0.049	0.99
Fixed	Π	$0b$	0.695	0.723	1.04
Fixed	Π	$1b$	0.175	0.111	0.63
Fixed	Π	$2b$	0.074	0.070	0.95

Table 4: Precision of each ten-replica mean. Native MC errors propagate the saved integration errors; replica standard errors use the scatter of the ten central estimates. Their ratio is a dispersion diagnostic, not a formal goodness-of-fit probability.

0.057%. For two tags these ranges are 0.053–0.074% and 0.061–0.070%, respectively. The least precise integrated selection is the multiplicative zero-tag rate, with approximately 0.70% native and 0.72% replica error. Across the entries in Table 4, the ratio of empirical to native errors ranges from 0.61 to 1.33. There is no large dispersion excess in these integrated diagnostics, although ten replicas cannot exclude a poorly sampled contribution in a rare bin.

Width-mode partners use strongly shared random streams and must also be compared as paired data. For the selected totals, fixed minus varying gives $(9.7 \pm 12.3) \times 10^{-6}$ fb for S and $(3.5 \pm 2.9) \times 10^{-6}$ fb for Π, using the empirical standard errors of paired differences. Both are consistent with their identical central definitions. The width convention changes the scale response, not the nominal physical input widths.

5 Differential comparison

We retain the useful observable families from the earlier report: charge-resolved lepton transverse momenta, positron pseudorapidity, separate bottom and antibottom jet spectra, lepton–jet invariant mass and angular separation, trilepton mass and same-sign azimuthal separation. Two-tag plots provide the most precise differential comparison; zero- and one-tag muon spectra expose category dependence. All figures now use the forty new integrations exclusively.

The absolute panels show native-bin densities and propagated native MC bars. Light shading is I63 and darker shading its C7 subset. Ratio panels show the ratio of mean central predictions, empirical paired jackknife MC bars, and scale-matched envelopes. The two width columns use the same displayed ranges and ratio-axis limits. Bins are not smoothed or merged. Native bins that straddle a cut remain native bins, and the density normalization uses their full booked widths.

Across the displayed two-tag spectra, median native relative bin errors are approximately 0.33–0.56% for S and 0.58–1.13% for Π. The selected zero-tag muon spectrum has medians near 1.26% and 2.28%. These summaries depend on the displayed ranges; individual tails remain less precise. The complete catalogue and displayed-bin precision inventories accompany the figures.

One remaining limitation is visible near $\Delta R(\mu^+, j_b) = 3$. In the 2.8–3.0 and 3.0–3.2 bins, the varying-width ratios are 0.865 ± 0.085 and 1.058 ± 0.088 from the paired replica estimate. The multiplicative native bin errors are approximately 9.8% and 8.3%, despite the much smaller integrated error. These excursions should not be interpreted as a resolved fine angular structure. The fixed-width partners show the same sampling fluctuations; they are not an independent confirmation. We retain the bins and their errors.

The correct-charge $m(e^+, j_b)$ observable illustrates a clear shape dependence. Its on-shell Born endpoint is $\sqrt{m_t^2 - m_W^2} \simeq 152.6$ GeV. The combined ratios are about 0.956 in 90–120 GeV, 0.910 in 120–150 GeV, and 0.825 in the 150–180 GeV bin straddling that endpoint. Their paired statistical errors are approximately 0.0021, 0.0015 and 0.0083, respectively. The decrease near the endpoint is much larger than the integrated four-percent normalization change. Radiation and jet contamination can populate masses above the Born boundary; those bins probe different perturbative content from the Born-populated region. Neither a resummation nor a complete NNLO result is supplied by S or Π.

$m(e^+, j_b)$ [GeV]	Width	Π/S	I63 ratio range	MC S [%]	MC Π [%]	MC Π/S
90–120	Varying	0.9563	$[0.9433, 0.9691]$	0.10	0.18	0.00207
90–120	Fixed	0.9563	$[0.9362, 0.9731]$	0.10	0.18	0.00204
120–150	Varying	0.9103	$[0.8824, 0.9376]$	0.09	0.19	0.00146
120–150	Fixed	0.9105	$[0.8767, 0.9407]$	0.09	0.19	0.00148
150–180	Varying	0.8250	$[0.7784, 0.8615]$	0.40	0.95	0.00834
150–180	Fixed	0.8254	$[0.7731, 0.8697]$	0.40	0.95	0.00828

Table 5: Selected native two-tag invariant-mass bins. MC S and MC Π are relative propagated errors in percent; MC Π/S is the absolute paired jackknife standard error of the dimensionless ratio. The I63 interval is a scale envelope, not a statistical interval.

6 Relation to full off-shell and higher-order calculations

The smaller correlated fixed-width band should be described as cancellation-sensitive, not as a demonstrated underestimate of a known true uncertainty. Published full off-shell NLO calculations also use fixed top widths and common scale variations. In Ref. [3], the full off-shell and full NWA rate bands are approximately $^{+5\%}_{-7\%}$ for $m_t + m_W/2$ and $^{+3\%}_{-6\%}$ for $H_T/3$. With LO decays the bands are instead about 10–11%; production–decay compensation is explicitly observed. The authors estimate a possible increase of up to about 1.5 percentage points when allowing the width to vary in their setup. Their integrated off-shell–NWA difference is below 0.2% with unexpanded width factors; expanding those normalizations changes the NWA result by 3–4%. These are different cuts, scales and conventions from the present study, so they give context rather than a matched validation.

An off-shell NLO comparison tests finite-width, non-resonant and non-factorizable effects; it does not supply missing factorizable NNLO corrections. Small integrated off-shell effects also do not guarantee small differential ones. Ref. [4] finds differences of roughly 27–35% near $p_T(b_1) = 600$ GeV, where single-resonant contributions become important. A wider scale envelope within our on-shell calculation cannot by itself estimate these omitted terms.

Higher-order production predictions provide a distinct calibration. Ref. [5] finds the inclusive, charge-summed NLO QCD rate 622.0 fb with a +12.7% upper scale variation, followed by an NNLO rate of 710.7 fb, a roughly 14% increase. Its finite two-loop term is approximated with a separate uncertainty. A direct leading-colour two-loop calculation subsequently confirms a sizeable positive NNLO effect, while retaining its own subleading-colour uncertainty [6]. Neither inclusive correction can simply be transferred to the present fiducial bins.

For a paper-level comparison, both C7 and independent production/decay envelopes should therefore be shown and the width convention stated. I63 is useful as a conservative sensitivity test, not a calibrated probability distribution. A matched full off-shell comparison should control the cuts, PDFs, scales

and width-expansion convention before attributing discrepancies to finite-width physics. The observed S–Π spread remains a separate selected-higher-order prescription comparison rather than a complete theory-error estimate.

7 Figures

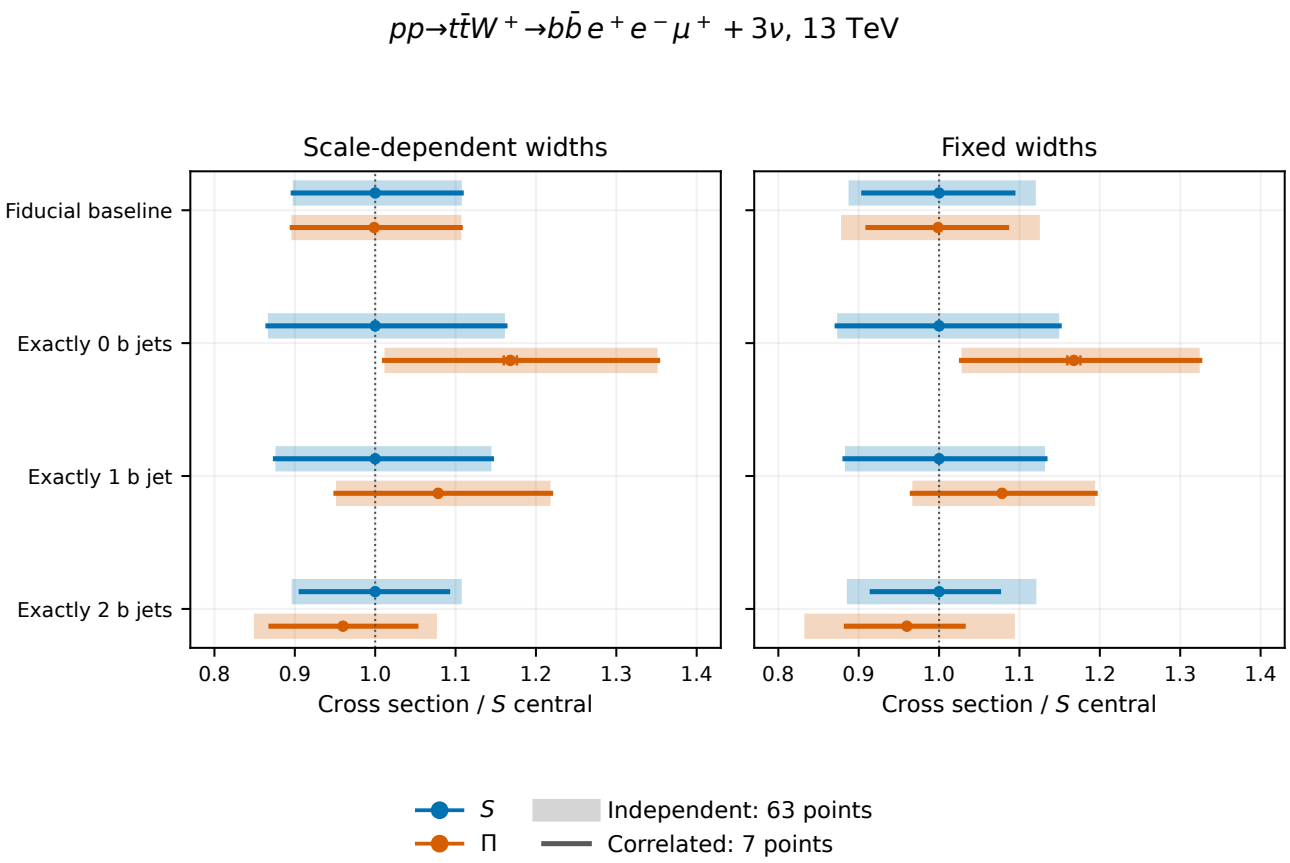


Figure 1: Baseline-selected total and exclusive tag rates. Each horizontal unit is the S mean central value in that selection and width convention. Thick light intervals show I63 and thin solid intervals C7. Markers carry propagated native MC errors. The plotting reference does not assign an independent error to a ratio.

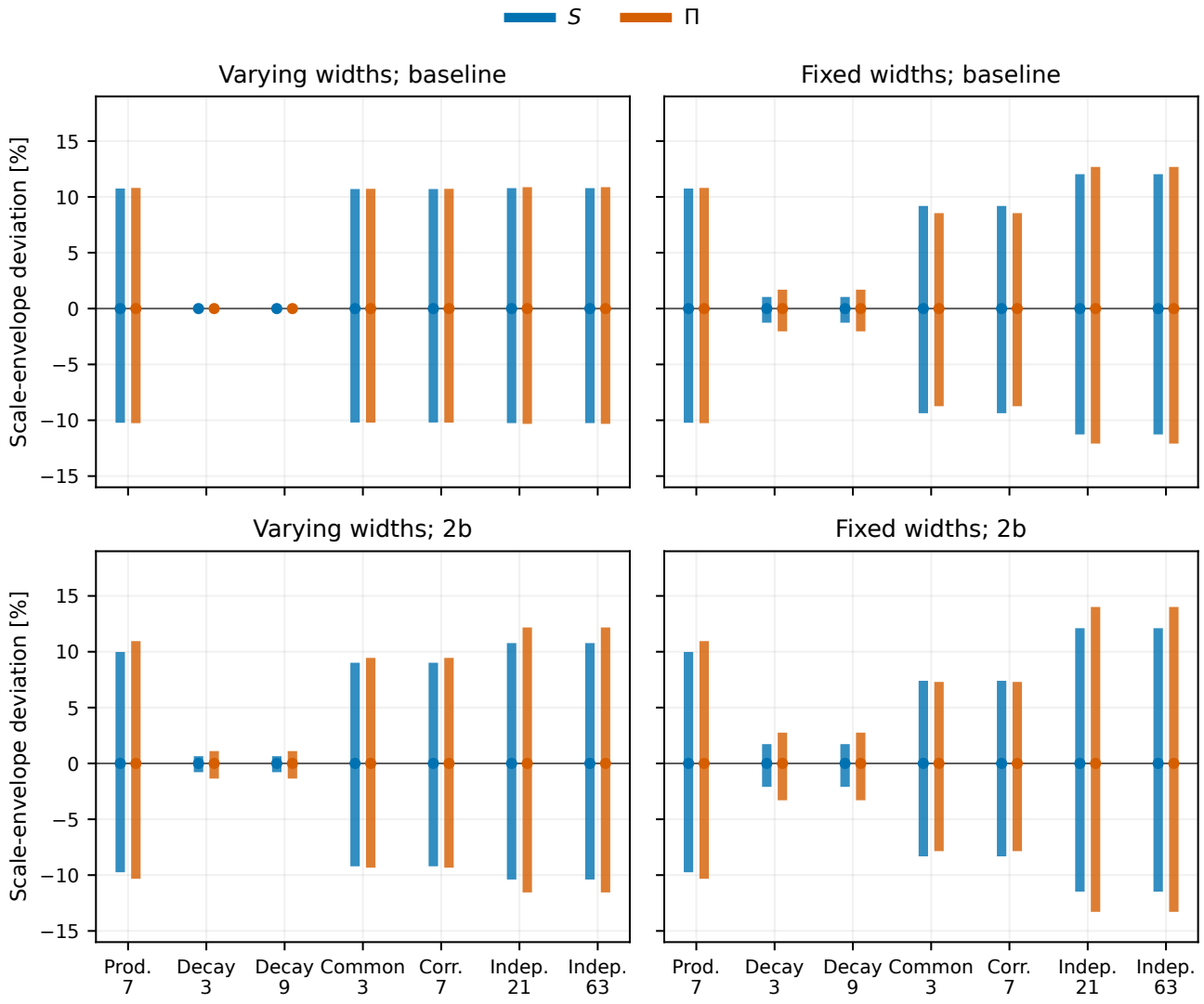


Figure 2: Alternative scale envelopes for the baseline total and two-tag rate. Each interval is measured relative to its own central prediction; markers sit at zero. Freezing widths strengthens the opposing decay response and reduces the envelope when scales are tied. The separate decay axes do not enlarge these particular integrated extrema.

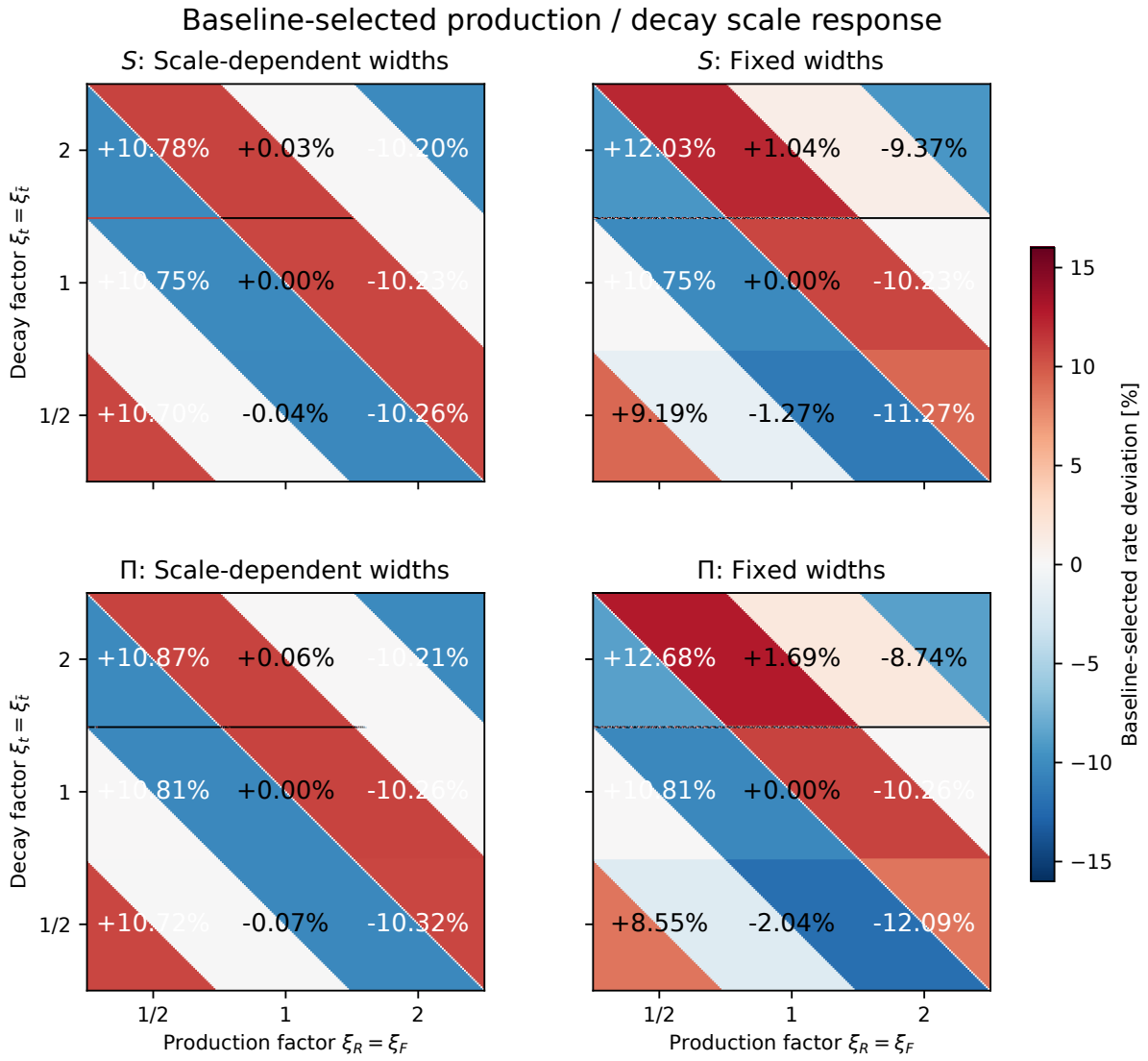


Figure 3: Baseline-selected production/decay response slices, with $\xi_R = \xi_F$ and $\xi_t = \xi_{\bar{t}}$. Cells give deviations from the corresponding central mean. Common3 follows the diagonal; independent variations also reach off-diagonal cells. The response is computed from the averaged scale weights and is not a covariance or likelihood.

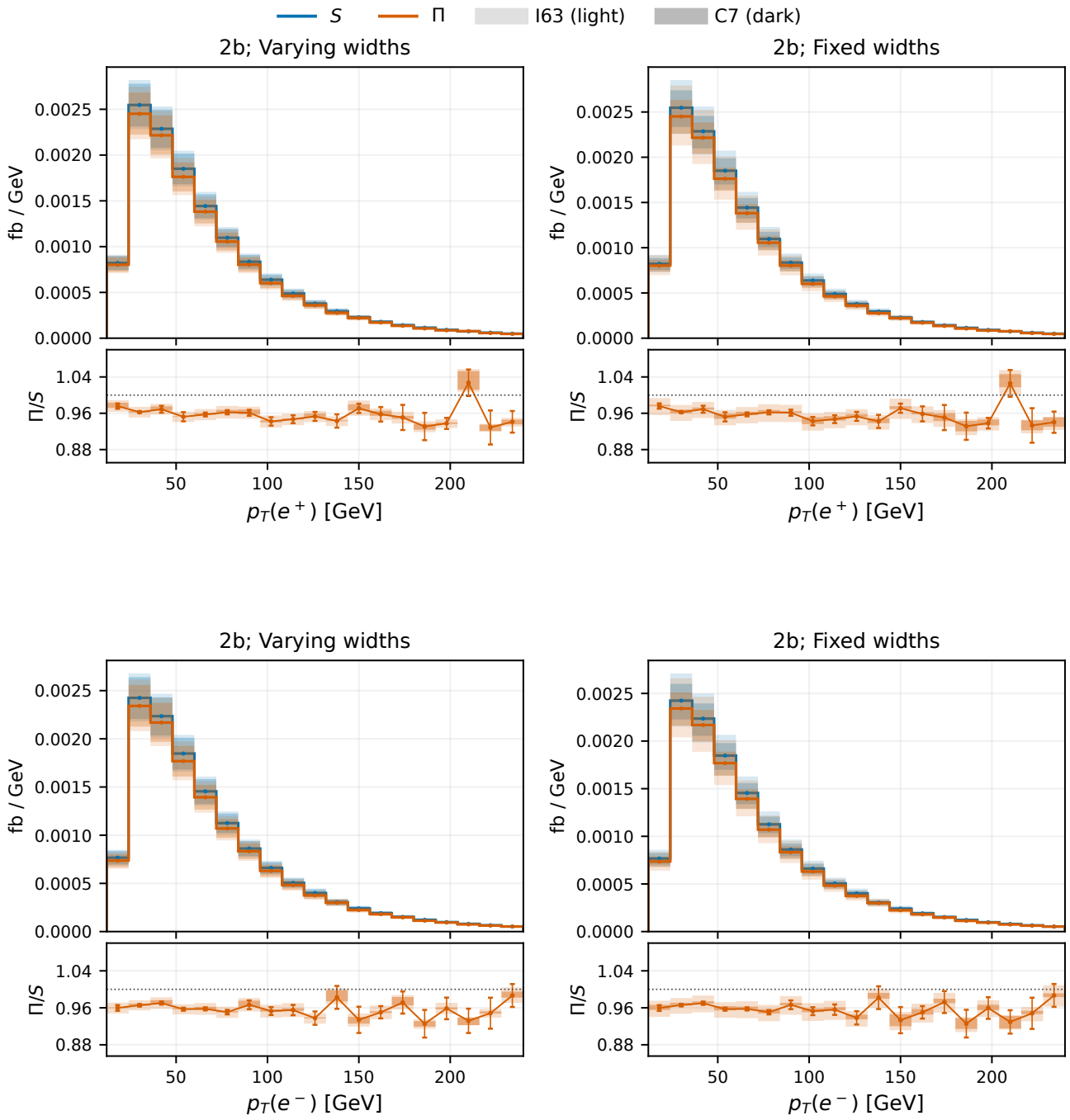


Figure 4: Two-tag electron and positron transverse momenta. Left: scale-dependent widths; right: fixed widths. Absolute densities carry propagated native MC bars, light I63 and darker C7 shading. Ratio panels carry paired-replica jackknife MC bars and point-matched scale envelopes. Native bins are retained, including bins straddling the 20 GeV lepton cut.

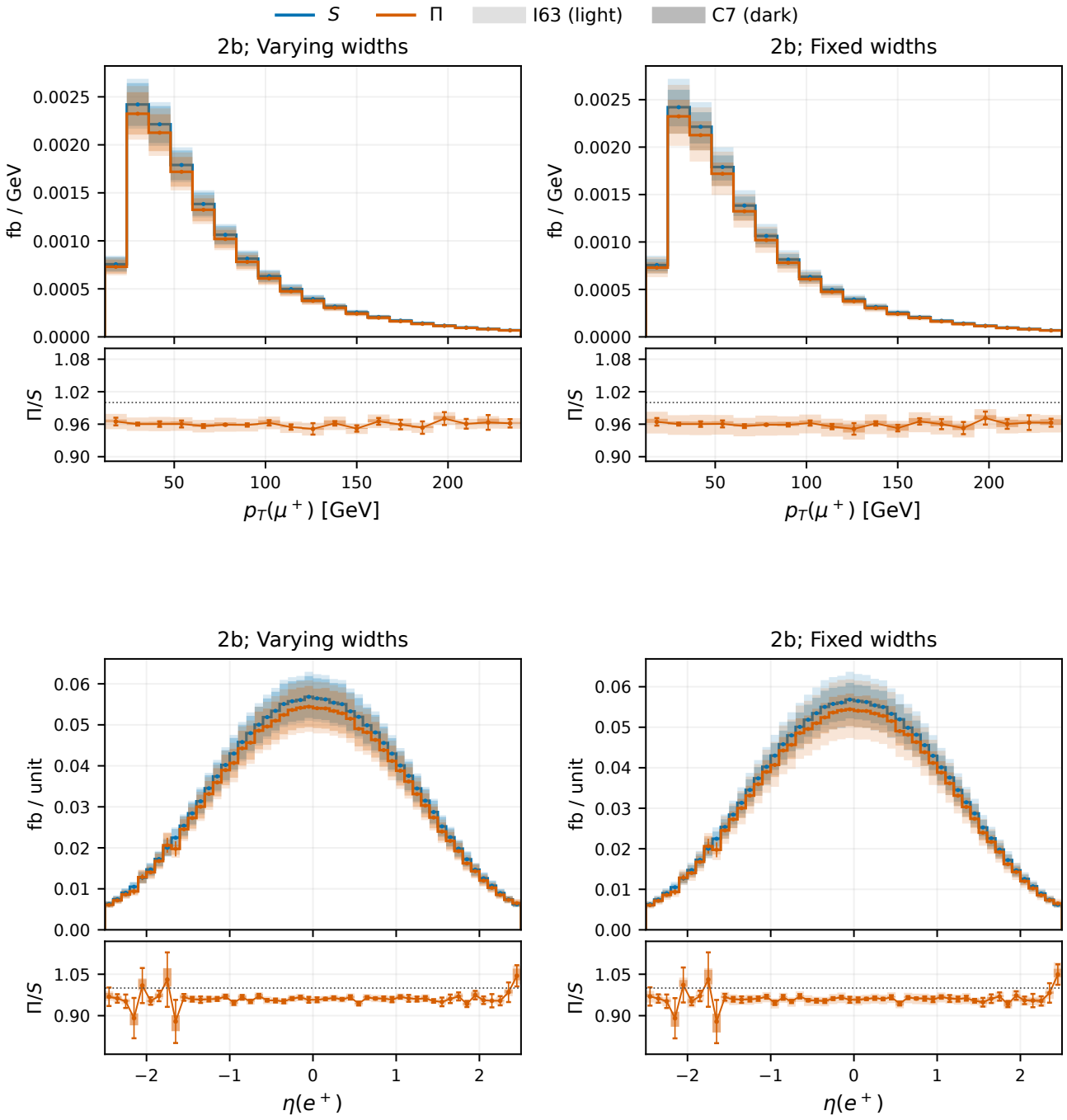


Figure 5: Two-tag associated-muon transverse momentum and positron pseudorapidity, with the conventions of Fig. 4. The muon probes production and spin information; the top-decay positron also responds directly to decay acceptance. Every charged lepton has the same $p_{\mathrm{T}} \geq 20$ GeV and $|\eta| < 2.5$ acceptance.

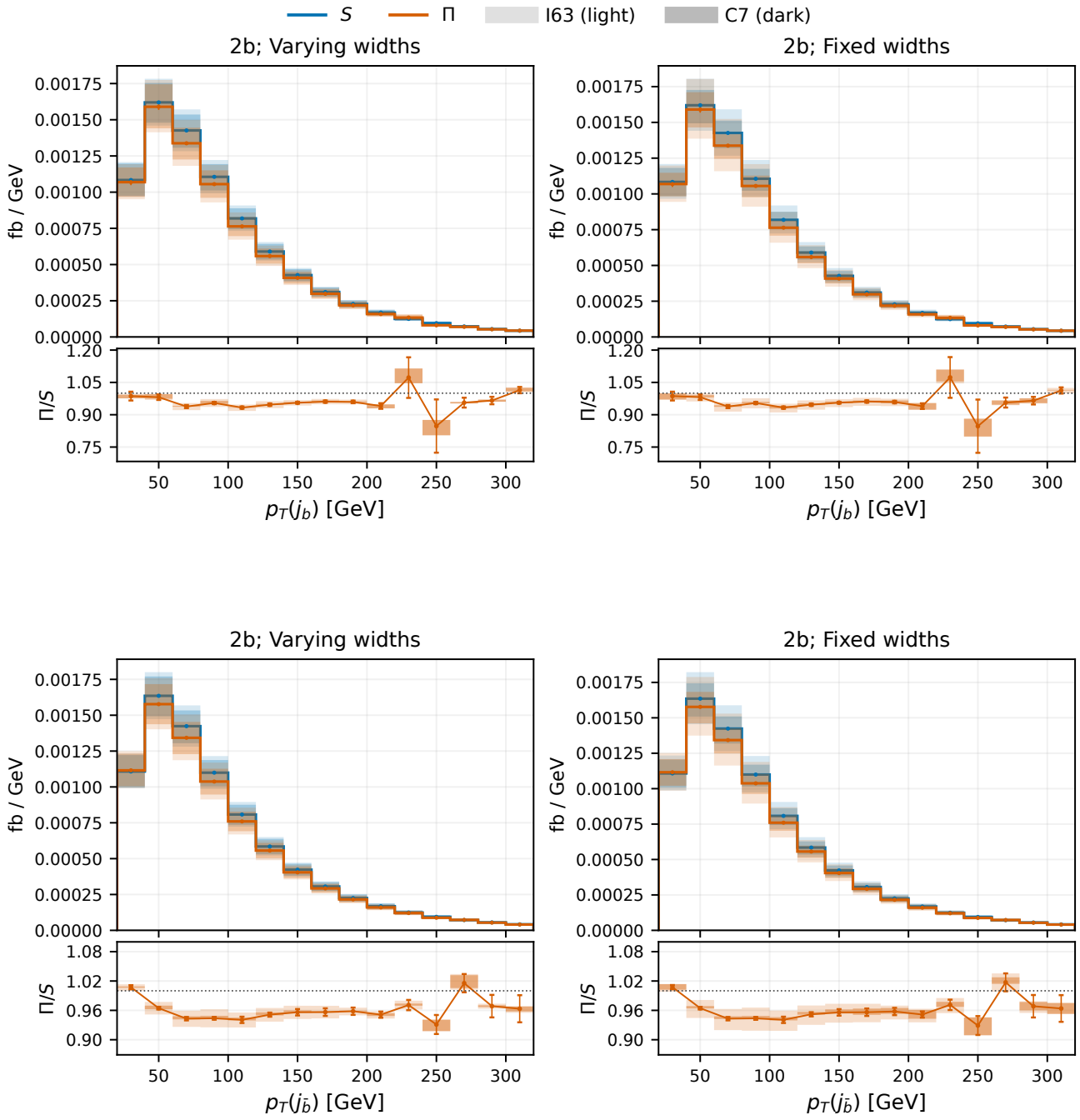


Figure 6: Separate accepted bottom- and antibottom-jet transverse momenta in the two-tag category. Jets containing both bottom flavours do not enter these distributions. Uniform lepton and jet acceptance does not impose identical spectra in the charge-asymmetric W^+ process. The densities are not shape-normalized.

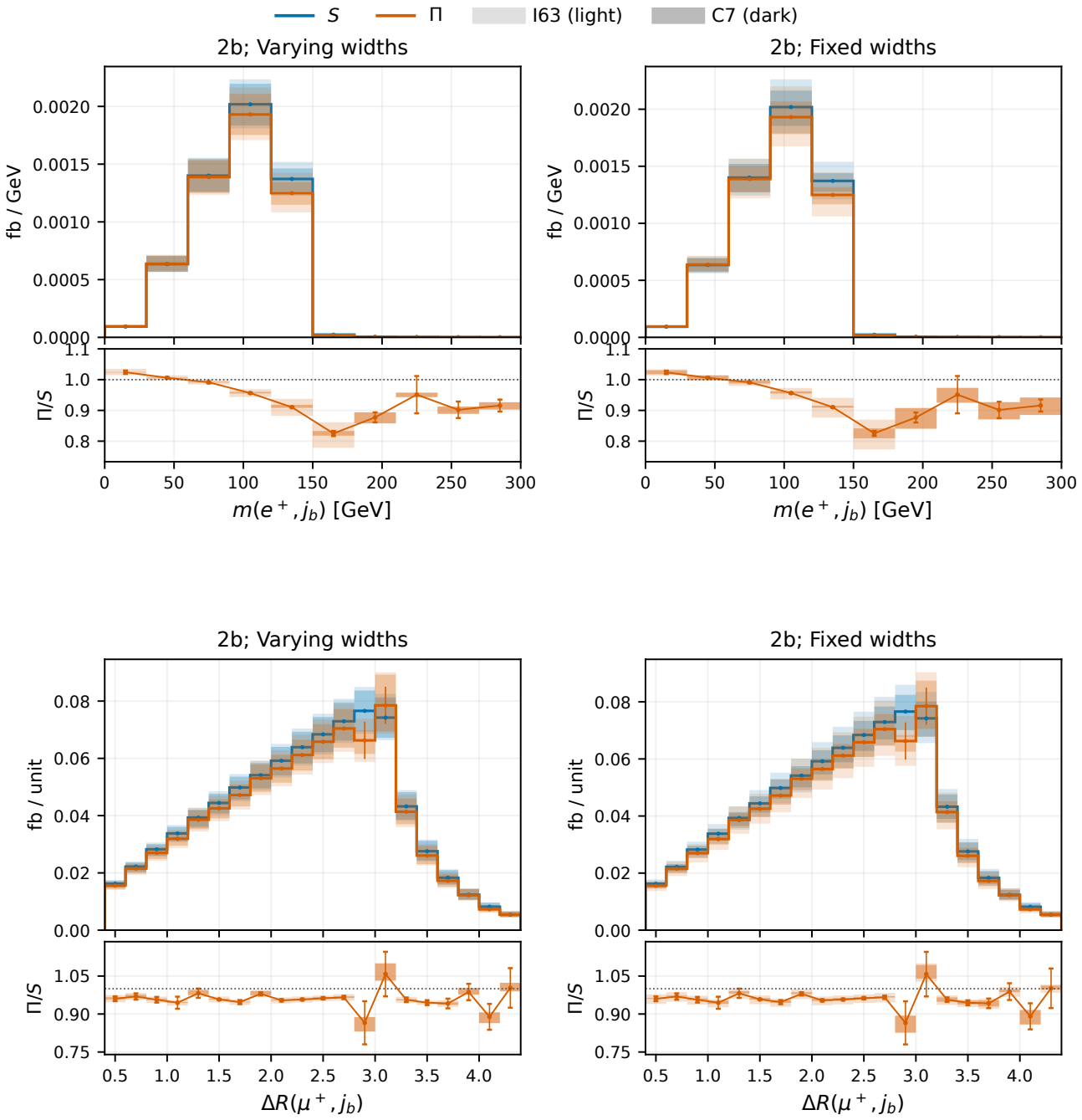


Figure 7: Two-tag $m(e^+, j_b)$ and $\Delta R(\mu^+, j_b)$, with absolute and ratio uncertainties as in Fig. 4. The mass uses the top-decay positron and the jet containing its bottom parton in this direct-lepton assignment. Jet radiation can populate bins beyond the approximately 152.6 GeV Born endpoint. The angular combination supplies a complementary directly visible correlation.

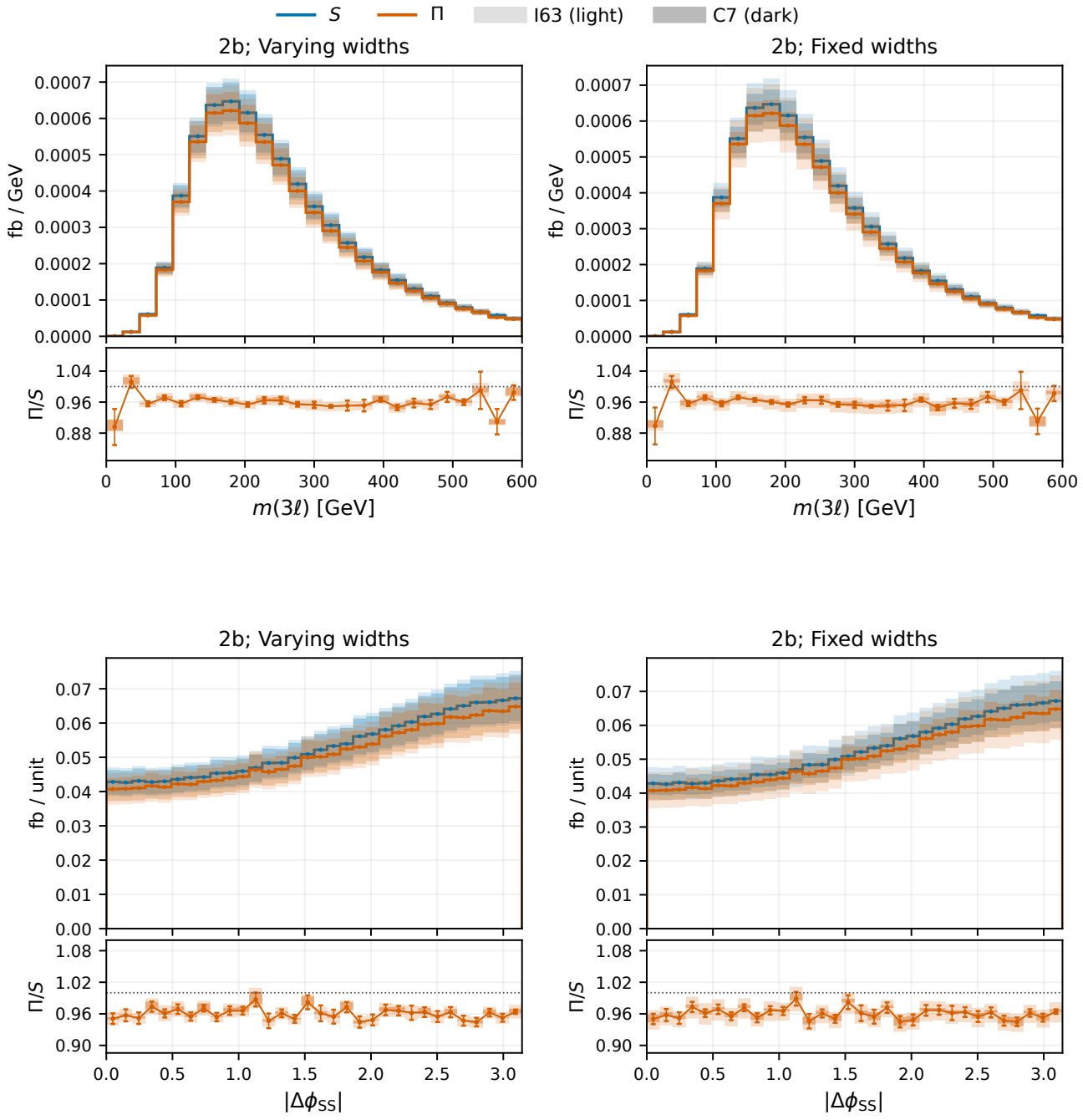


Figure 8: Two-tag trilepton invariant mass and same-sign azimuthal separation. Here the same-sign pair is $e^+\mu^+$. There is no Z-mass veto. These combinations require no neutrino fit or reconstructed parent assignment. Scale shading and paired statistical bars are distinct uncertainty estimates.

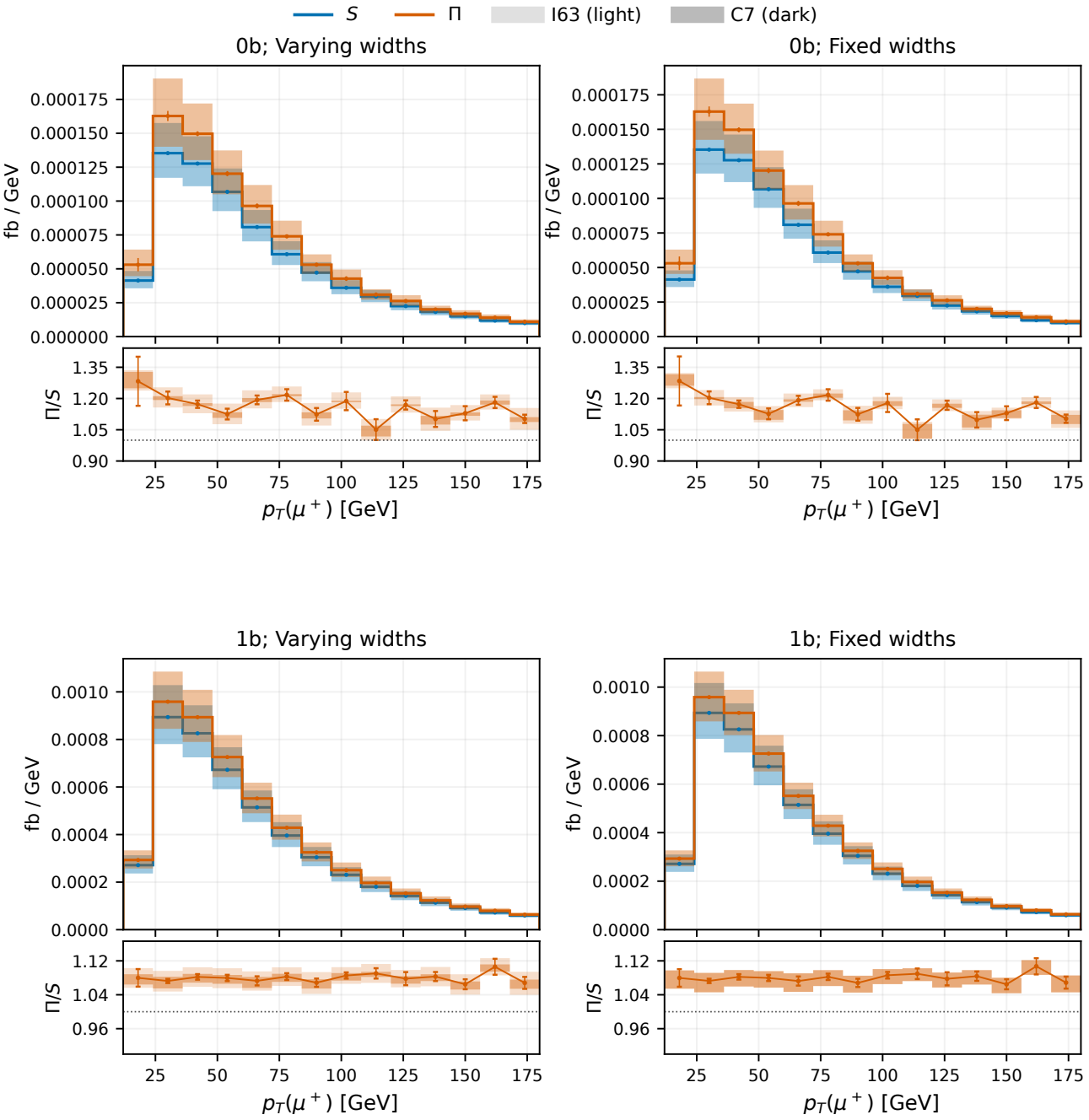


Figure 9: Associated-muon transverse momentum in the exclusive zero-tag and one-tag categories. Their multiplicative enhancements are visible at substantially improved precision. Rare zero-tag bins remain less precise; no smoothing or removal of populated native bins is used. Ratio bars estimate MC precision from paired replicas and must not be confused with the shaded scale envelopes.

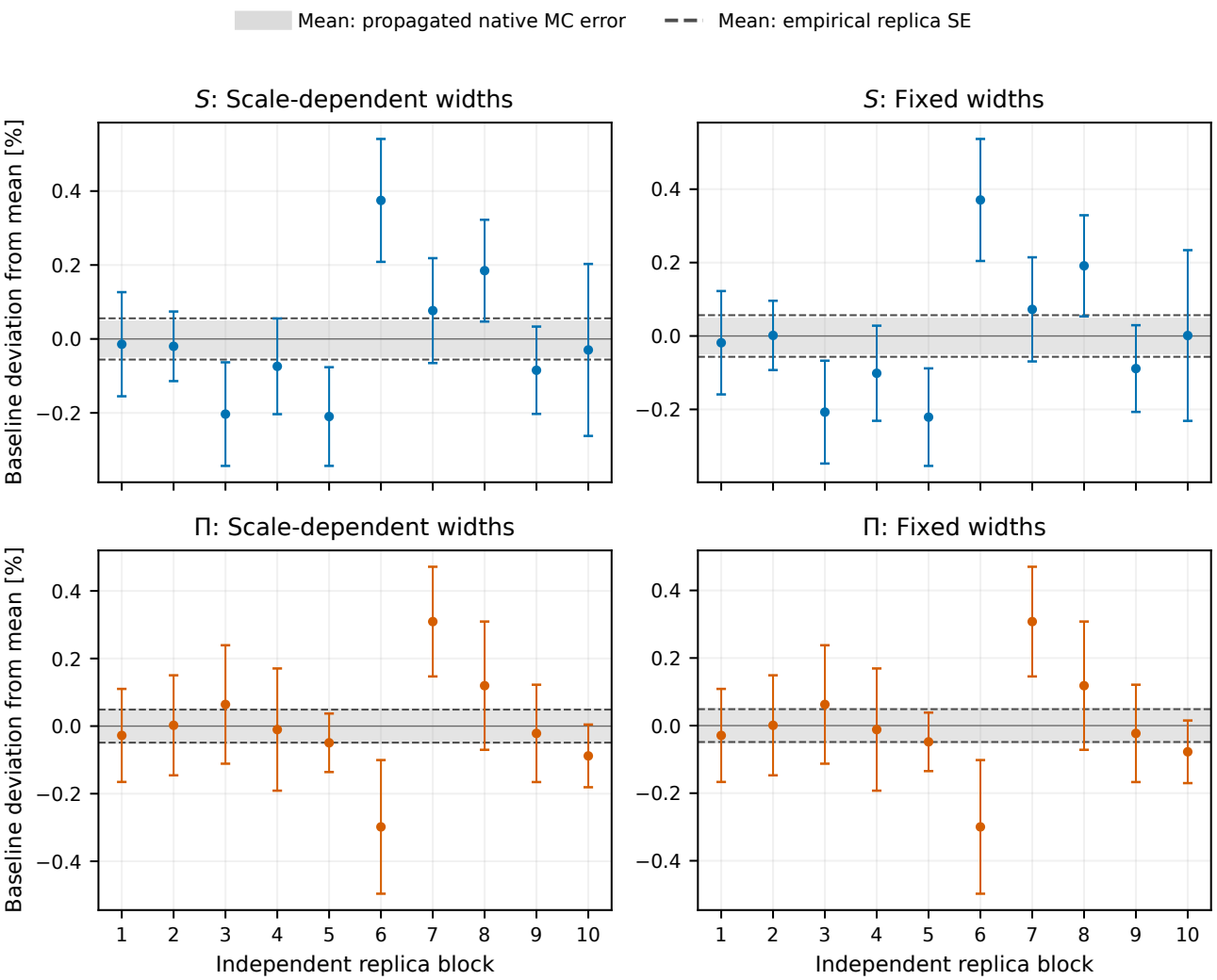


Figure 10: Baseline rate in each repeated integration, relative to its ten-replica mean. Bars show the native error of each integration. The grey interval is the propagated native error on the mean and dashed lines show the empirical standard error on the mean. Width partners share most streams, so these panels are not four independent ensembles for cross-prediction comparisons.

A Reproducibility and scope

The report directory contains this LaTeX source and PDF, five generated tables, ten PDF/PNG figures, archived cards for all forty integrations, installed source snapshots, the numerical `results.json`, the combined `collected.npz`, figure and report manifests, and the full-catalogue precision CSV. The combined arrays retain the central and native error of every replica in every bin, the empirical error on each mean, and all 81 mean scale weights. Original run files are read-only inputs. Their paths and hashes are recorded.

Collection verifies complete compatible histogram layouts and scale coordinates, matched reference cards, fixed-width inputs, the central scale weight, and tag-category closure. It also checks that different replica blocks have no shared logged seed pairs. These are consistency checks and a sampling audit, not a proof of convergence in every tail. The combined sample retains native bins: no unsupported within-run quadrature rebinning or artificial smoothing is used. The signed decay labels print `d-6` before `d6`; the reduction explicitly orders coordinates as production μ_R, production μ_F, top μ_R, antitop μ_R.

From the repository root, reproduce the reduction and figures with

```
python ttw_study/scripts/collect_decay_scale_report.py \
  --campaign-root . --output ttw_study/reports/decay_scales_20261030
python ttw_study/scripts/render_decay_scale_report.py \
  --output ttw_study/reports/decay_scales_20261030
cd ttw_study/reports/decay_scales_20261030
make
```

The source ZIP contains the document, generated table inputs, standalone PDF figures and build instructions, sufficient to rebuild the report. The larger numerical arrays and original HwU files are not required merely to compile that ZIP. The report prose is specific to these inputs and should be reviewed if the campaign is changed.

This sample does not include a direct-lepton-flavour sum, W^-, PDF uncertainty, non-factorizable or finite-width corrections, electroweak corrections, parton-shower matching or a complete NNLO calculation. Existing full off-shell results provide comparison targets under matched conditions; they have not been rerun with these cuts. The earlier neutrino-fit observables remain retired. Statistical fluctuations or endpoint sensitivity in the present visible observables do not by themselves establish a new infrared-safety failure.

References

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[2] R. Frederix, S. Frixione, F. Maltoni and T. Stelzer, *Automation of next-to-leading order computations in QCD: the FKS subtraction*, JHEP 10 (2009) 003, arXiv:0908.4272.

[3] G. Bevilacqua, H.-Y. Bi, H. B. Hartanto, M. Kraus and M. Worek, *The simplest of them all: $t\bar{t}W^{\pm}$ at NLO accuracy in QCD*, JHEP 08 (2020) 043, arXiv:2005.09427.

[4] G. Bevilacqua et al., *Modeling uncertainties of $t\bar{t}W^{\pm}$ multilepton signatures*, Phys. Rev. D 105 (2022) 014018, arXiv:2109.15181.

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[6] M. Becchetti et al., *NNLO QCD predictions for $t\bar{t}W$ production at hadron colliders*, arXiv:2606.09503.